

Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions through data-driven analysis of spending patterns, customer segments, and product preferences.



Dataset Overview

3,900

Total Purchases

Transactions analyzed across
all categories

18

Data Columns

Comprehensive customer
and purchase attributes

\$60

Average Purchase

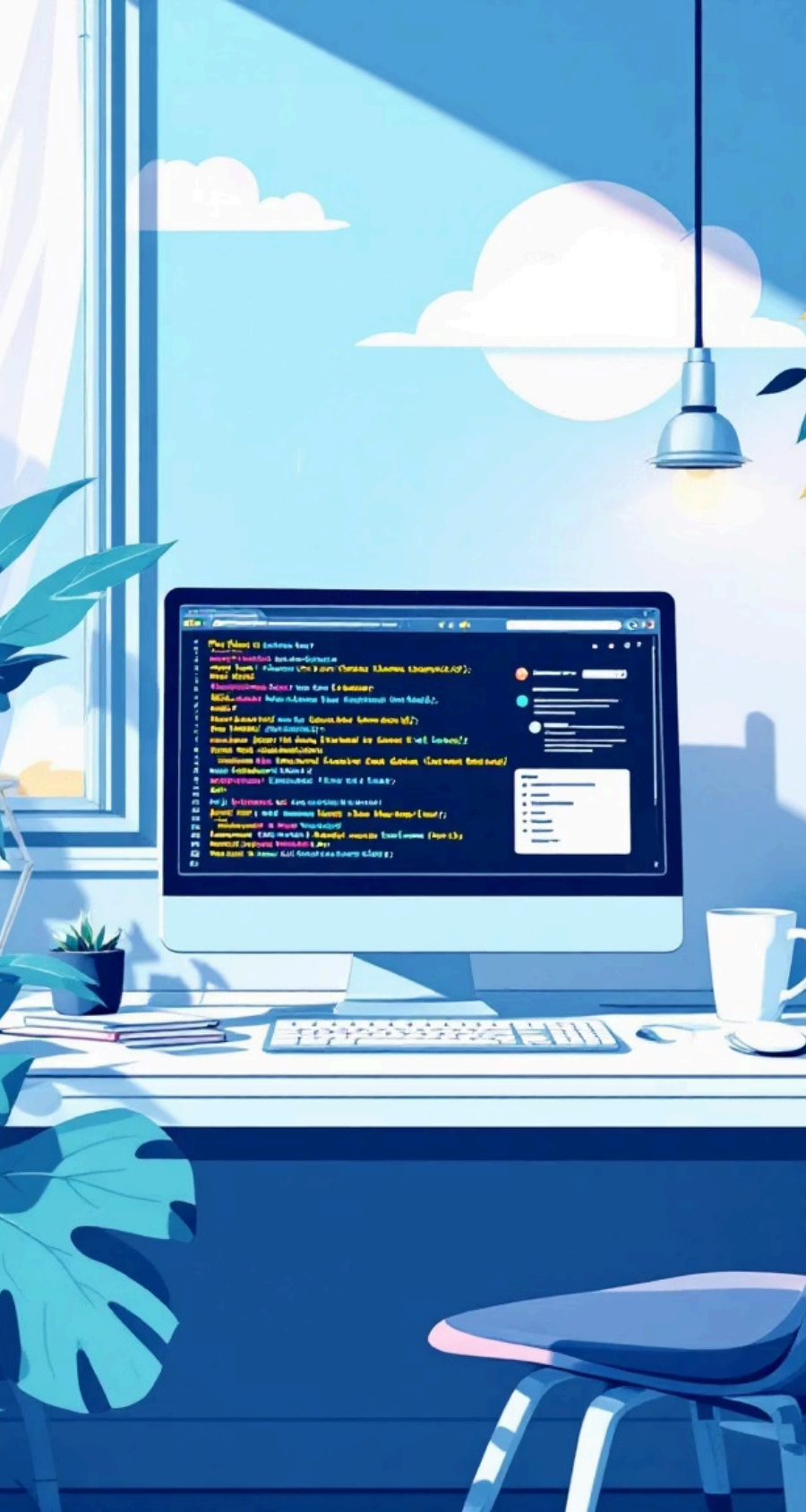
Mean transaction value
across all customers

50

Locations

Geographic distribution of
customer base

Dataset includes customer demographics, purchase details, shopping behavior metrics, and subscription status with minimal missing data (37 values in Review Rating).



Data Preparation Journey

01

Data Loading & Exploration

Imported dataset using pandas and performed initial structure analysis with `df.info()` and summary statistics.

02

Cleaning & Standardization

Handled missing values, renamed columns to snake_case, and verified data consistency.

03

Feature Engineering

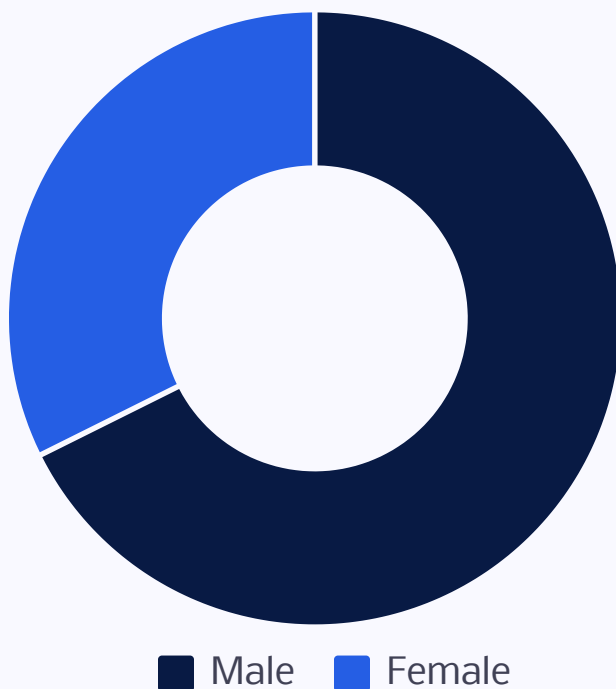
Created `age_group` bins and `purchase_frequency_days` columns for deeper analysis.

04

Database Integration

Connected to PostgreSQL and loaded cleaned data for SQL-based business analysis.

Revenue Insights by Gender

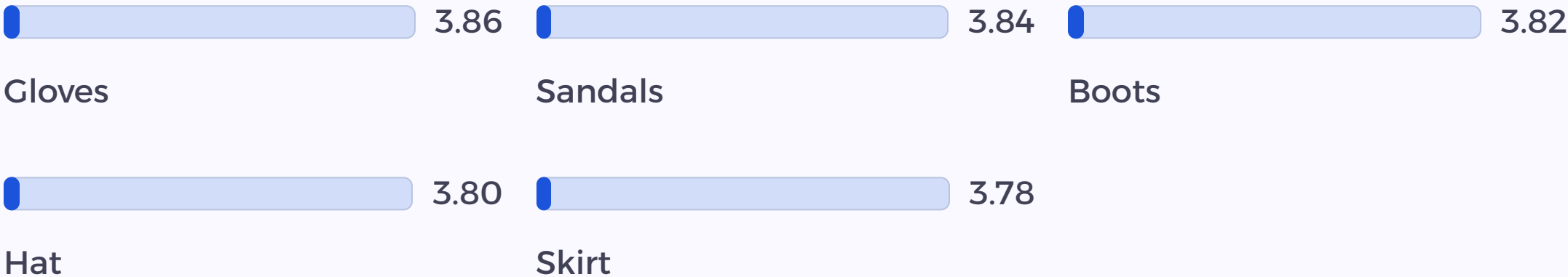


Male Customers Drive Revenue

Male customers generated **\$157,890** compared to **\$75,191** from female customers, representing 68% of total revenue.

This significant gap suggests opportunities for targeted marketing to increase female customer engagement.

Top-Rated Products



Average review ratings reveal customer satisfaction leaders. These products should be highlighted in marketing campaigns and featured prominently in product positioning strategies.

Customer Segmentation Analysis

Loyal Customers

3,116 customers (80% of base)

Highest value segment with consistent purchase history

Returning Customers

701 customers (18% of base)

Growth opportunity to convert into loyal segment

New Customers

83 customers (2% of base)

Fresh acquisition requiring engagement strategies

Strong loyal customer base indicates effective retention, but new customer acquisition needs attention.

Subscription Impact

Subscribers vs. Non-Subscribers

- **1,053 subscribers** with \$59.49 average spend
- **2,847 non-subscribers** with \$59.87 average spend
- Similar spending patterns across both groups
- Repeat buyers (>5 purchases): 958 subscribers vs. 2,518 non-subscribers



❏ **Key Finding:** Subscription status doesn't significantly impact average spend, suggesting untapped potential for exclusive subscriber benefits to drive differentiation.



Discount Strategy Insights

1

High-Spending Discount Users

839 customers used discounts but still spent above average (\$60+), indicating price sensitivity doesn't always mean low value.

2

Discount-Dependent Products

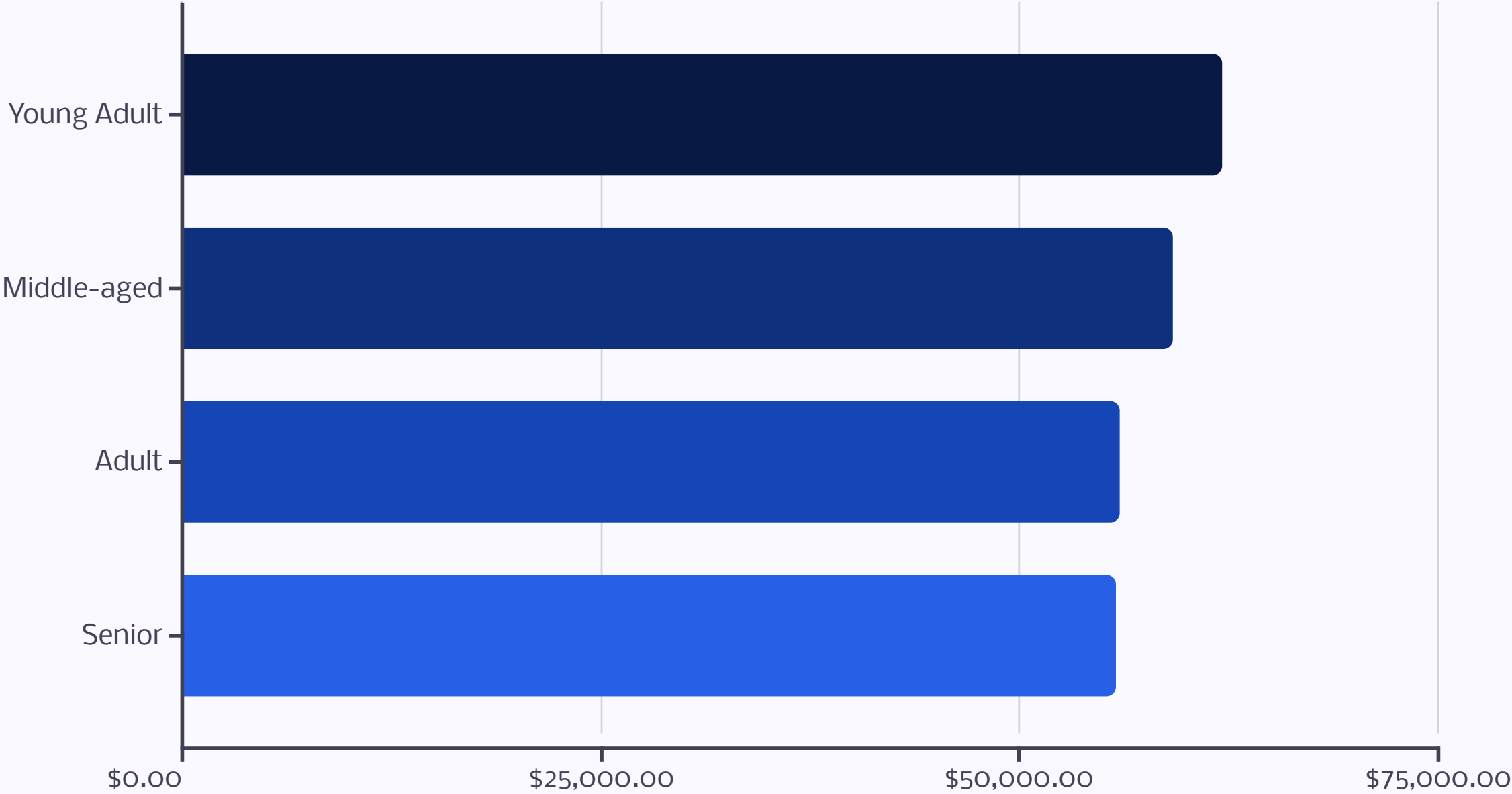
Hat (50%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), Pants (47.37%) show highest discount rates.

3

Strategic Balance

Need to balance promotional sales boosts with margin control to optimize profitability.

Revenue by Age Group



Young adults lead revenue generation at **\$62,143**, with relatively balanced contribution across all age groups. Express shipping users show \$60.48 average spend vs. \$58.46 for standard shipping.

Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits and perks to differentiate subscriber experience and increase conversion.



Loyalty Programs

Reward repeat buyers to accelerate movement into loyal customer segment.



Targeted Marketing

Focus on high-revenue age groups and express-shipping users for maximum ROI.



Product Positioning

Highlight top-rated and best-selling products in campaigns to drive conversions.



Discount Optimization

Balance promotional sales with margin control for sustainable profitability.